

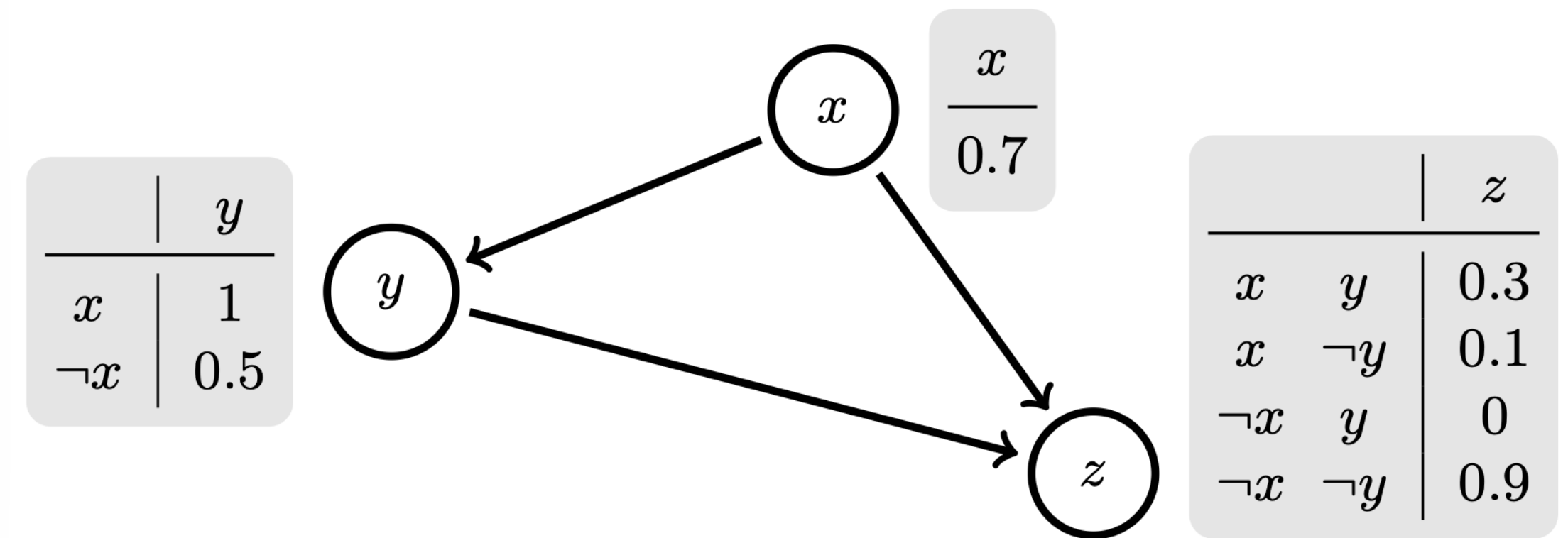
Probabilistic Ontologies

Bayesian Description Logic: $\mathcal{BEL}$

Probabilistic GCI: $\langle C \sqsubseteq D : x \rangle$

Crisp (classical) GCI: $\langle C \sqsubseteq D \rangle$

Bayesian Network



Contexts determine necessary conditions:

$\langle \text{LuxuryHotel} \sqsubseteq \exists \text{hasFeature.MeetingRoom} : \text{city} \rangle$

$\langle \text{LuxuryHotel} \sqsubseteq \exists \text{hasFeature.SkiRentalService} : \text{ski_resort} \rangle$

OWL2 DL: $\mathcal{SRQI\mathcal{Q}(\mathcal{D})}$

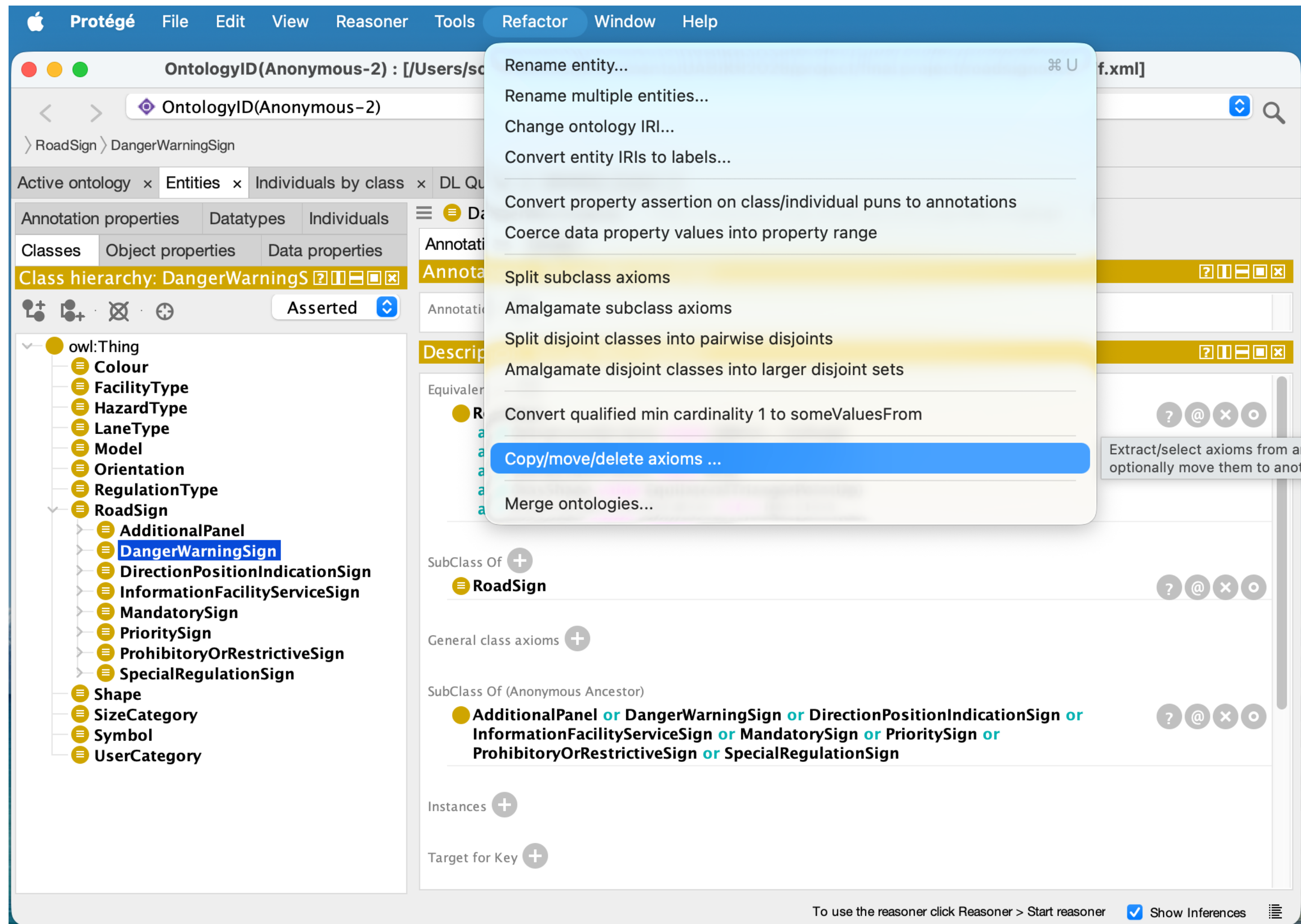
Feature	DL Syntax	Example / Description
Atomic Concept	A	A named class (e.g., Person).
Top / Bottom	$\top, \perp$	The set of all individuals / the empty set.
Intersection	$C \sqcap D$	Things that are both C and D .
Union	$C \sqcup D$	Things that are either C or D (or both).
Negation	$\neg C$	Things that are not C .
Existential Restriction	$\exists R.C$	Has at least one R -relationship to C .
Universal Restriction	$\forall R.C$	All R -relationships must be to C .
Nominals ($\mathcal{O}$)	$\{a_1, \dots, a_n\}$	A class defined by specific individuals.
Qualified Cardinality ($\mathcal{Q}$)	$\geq n R.C, \leq n R.C$	At least/most n relationships to class C .
Inverse Roles ($\mathcal{I}$)	R^-	The reverse of a property (e.g., hasParent vs hasChild).
Role Inclusion ($\mathcal{R}$)	$R \sqsubseteq S$	If xRy , then xSy .
Complex Role Chains ($\mathcal{R}$)	$R_1 \circ \dots \circ R_n \sqsubseteq S$	e.g., hasParent $\circ$ hasBrother $\sqsubseteq$ hasUncle.
Self-Restriction ($\mathcal{S}$)	$\exists R.Self$	Individuals related to themselves via R (e.g., Auto-regulating).
Transitivity ($\mathcal{S}$)	$Trans(R)$	If aRb and bRc , then aRc .
Symmetry / Asymmetry	$Sym(R) / Asym(R)$	Defines bidirectional or strictly one-way links.
Reflexivity / Irreflexivity	$Ref(R) / Irref(R)$	Whether aRa is always true or always false.
Disjoint Roles	$Dis(R, S)$	R and S cannot share the same pair of individuals.
Datatypes ($\mathcal{D}$)	$D, \exists T.d, \dots$	Integration of concrete data values (strings, integers).

OWL2 EL: $\mathcal{EL}^{++}$

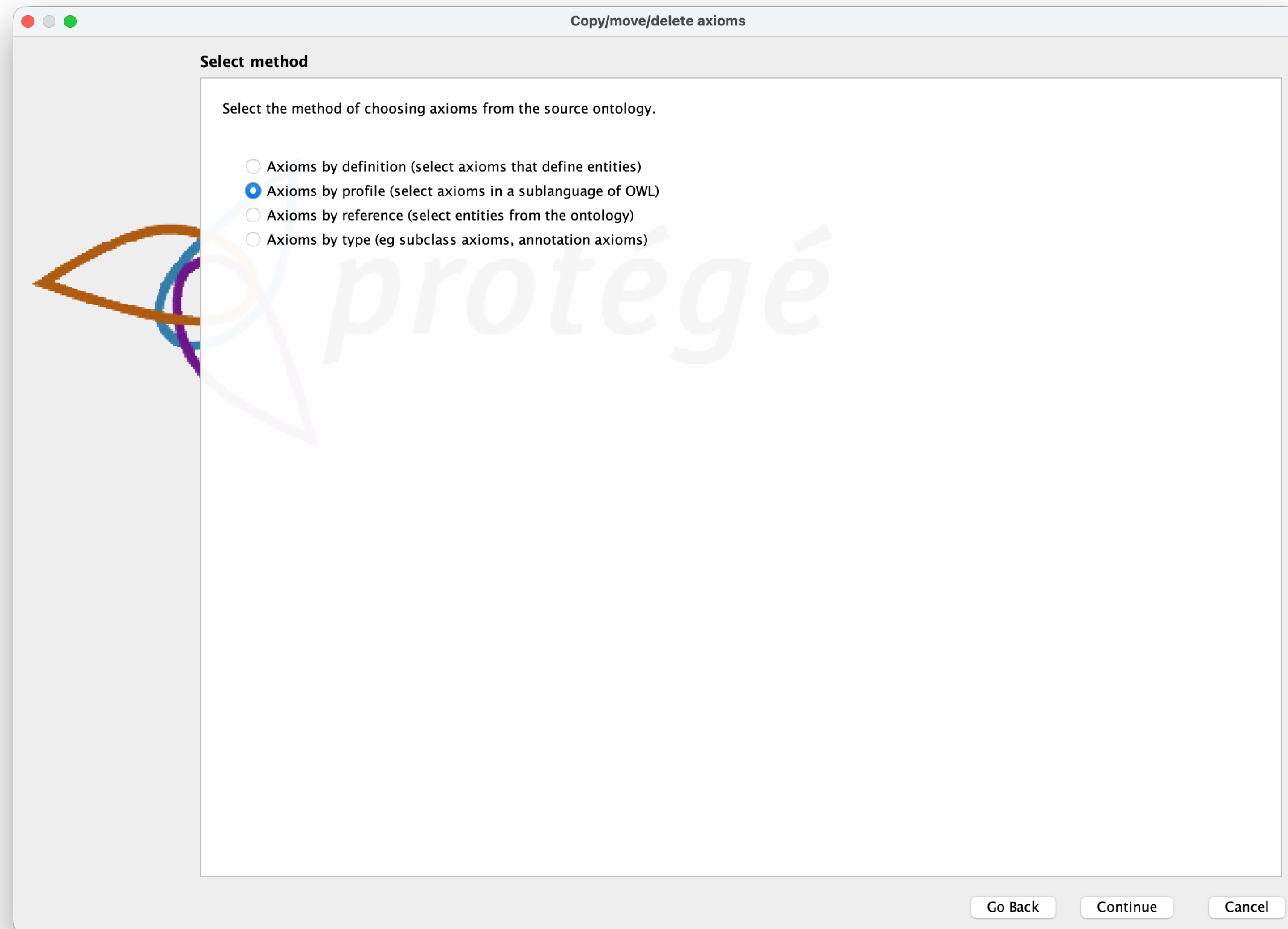
Feature	DL Syntax	Example / Description
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Existential Restriction	$\exists R.C$	Has at least one R -relationship to C .
Universal Restriction	$\forall R.C$	All R -relationships must be to C .
Nominals ($\mathcal{O}$)	$\{a\}$	A class defined by a single individual.
Qualified Cardinality ($\mathcal{Q}$)	$\geq n R.C, \leq n R.C$	At least/most n relationships to class C .
Inverse Roles ($\mathcal{I}$)	R^-	The reverse of a property (e.g., hasParent vs hasChild).
Role Inclusion ($\mathcal{R}$)	$R \sqsubseteq S$	If xRy , then xSy .
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$\mathcal{EL}$

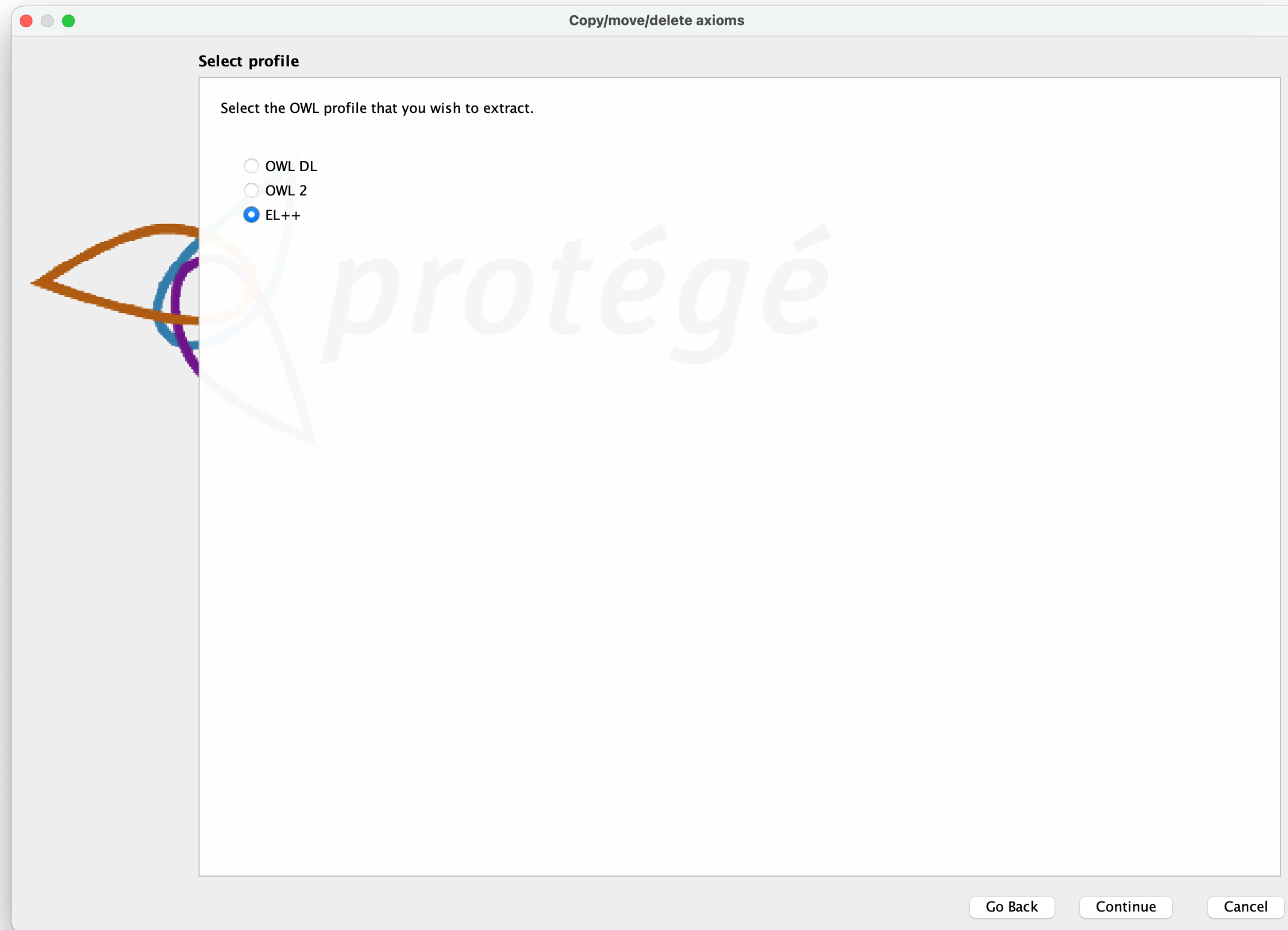
Extracting EL Profile



Extracting EL Profile



Extracting EL Profile



OWL2 EL Ontology ($\mathcal{EL}^{++}$)

untitled-ontology-397 (http://www.semanticweb.org/schorlemmer/ontologies/2026/4/untitled-ontology-397) : [http://www.semanticweb.org/schorlemmer/ontologies/2026/4/untitled-ontology-397]

untitled-ontology-397 (http://www.semanticweb.org/schorlemmer/ontologies/2026/4/untitled-ontology-397)

RoadSign > PrioritySign

Active ontology x Entities x Individuals by class x DL Query x SPARQL Query x

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: PrioritySign

- owl:Thing
 - Colour
 - FacilityType
 - HazardType
 - LaneType
 - Model
 - Orientation
 - RegulationType
 - RoadSign
 - AdditionalPanel
 - DangerWarningSign
 - DirectionPositionIndicationSign
 - InformationFacilityServiceSign
 - MandatorySign
 - PrioritySign**
 - ProhibitoryOrRestrictiveSign
 - SpecialRegulationSign
 - Shape
 - SizeCategory
 - Symbol
 - UserCategory

PrioritySign — http://example.org/roadsigns#PrioritySign

Annotations Usage

Annotations: PrioritySign

Annotations +

Description: PrioritySign

Equivalent To +

- RoadSign and (indicatesRegulationType value PriorityRule)

SubClass Of +

- RoadSign

General class axioms +

SubClass Of (Anonymous Ancestor)

Instances +

Target for Key +

Disjoint With +

- InformationFacilityServiceSign
- DirectionPositionIndicationSign
- MandatorySign
- DangerWarningSign

To use the reasoner click Reasoner > Start reasoner ☒ Show Inferences

Assignment

- Extract (or re-engineer) an $\mathcal{EL}$ -fragment of your road-sign ontology
- You may focus only on a fragment of the road-sign taxonomy (e.g., subclasses of ProhibitoryOrRestrictiveSign)
- Adjust your necessary conditions (GCIs) to different contexts

Contexts for Road Signs

European Speed Limit Sign



US Speed Limit Signs

